Venkata Lakshmi Parimala Pasupuleti

AI Solutions Architect · PropTech · Fintech · Renter Underwriting · Workflow Automation · Agentic AI · Regulatory Compliance · Python · 9yr Production

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

AI Solutions Architect with 9+ years of production engineering and 7 years of hands-on renter underwriting domain expertise - the exact intersection TheGuarantors is hiring for. At Entrata, I was the human layer in the income and identity verification workflow for multifamily affordable housing operators: reviewing income documentation, detecting misrepresentation, applying LIHTC/HCV/HUD income and asset limits, and flagging the same renter default risk that TheGuarantors' AI underwriting now automates at scale. I then encoded that manual judgment into production AI systems. LeaseI is a production 8-agent agentic workflow (LangGraph, FastAPI) that automates housing compliance investigation across document intelligence, regulatory rules, historical incidents, and live system data - pilot to production, sole architect. AgentProbe is the evaluation and monitoring infrastructure ensuring AI solutions stay reliable: 5-metric quality engine, auditability tracing, CI regression gate, and operational risk controls. At Magizhs, I was the sole technical owner embedding with an external client - translating ambiguous business requirements into a production AI system in weeks. At Entrata, I integrated with vendor fraud detection platforms (Snappt, Experian Precise ID) and delivered production systems to 100K+ housing units at 99.5% uptime under HUD/LIHTC/PHI compliance requirements. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Solutions & Architecture: Python (9yr), LangGraph, LangChain, MCP, Agentic AI, Multi-Agent Systems, AI-Enabled Workflows, Workflow Automation, Solutions Architecture, Pilot to Production, Vendor Integration, Claude API, OpenAI API, Anthropic API, Ollama, Prompt Engineering, Context Engineering, RAG Pipelines, Vector Databases, Semantic Search, FastAPI, REST APIs, API Design

Monitoring, Auditability & Compliance: Evaluation Frameworks, LLM-as-Judge, Hallucination Detection, Observability, Tracing, Monitoring, Auditability, CI Regression Gates, Data Quality, Data Governance, Operational Risk, Regulatory Compliance, LIHTC, HUD, HCV, PHI/HIPAA, RBAC, Security, Income Verification, Identity Fraud, Renter Underwriting, Lease Risk, Multifamily Operations

Infrastructure & Data: AWS (S3, EC2, RDS, Glue, Lambda), Docker, Microservices, RabbitMQ, PostgreSQL, MySQL, Oracle, Redis, GitHub Actions CI/CD, React.js, TypeScript, Tableau, Power BI, Python Automation, Linux, Git, Agile/Scrum, Cross-functional Delivery, Stakeholder Communication, Vendor Management

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - AI Solutions Delivery (External Client)

- Sole technical owner embedding with an external client - translated ambiguous operational requirements into a production AI solution (LangGraph, FastAPI, Python) from requirements through architecture, vendor evaluation, implementation, monitoring setup, and production deployment in weeks; wrote design docs; built HITL guardrails for regulated decision paths; delivered auditability, observability, and operational risk controls for ongoing operation. Pilot to production, sole architect.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); applied anomaly detection (Isolation Forest) surfacing data quality issues manual review missed; designed observability dashboards for faculty leadership (40% comprehension improvement); cross-functional delivery to non-technical stakeholders across 3 research pipelines.

Entrata - PropTech / Multifamily & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Renter underwriting domain - 7 years as the human income verification layer** - reviewed income documentation, detected income misrepresentation and fake pay stubs, applied LIHTC/HCV/HUD income and asset limits, flagged identity fraud and household composition discrepancies for multifamily affordable housing operators - the same renter default risk and income fraud TheGuarantors' AI underwriting automates. Entrata published industry research on this fraud problem. Automated 80% of this manual review workload via production pipelines with zero compliance violations.
- **Vendor integrations and solutions architecture** - integrated with Snappt (document fraud detection for pay stubs and bank statements), Experian Precise ID (identity verification), and HUD Upfront Income Verification APIs; owned the affordable housing compliance pipeline (180,000 domain records, 100K+ housing units) for 7 years at 99.5% uptime; AWS S3 document migration (100% PHI compliance, zero data loss); Docker microservices redesign (50% scalability improvement); RabbitMQ event-driven async with retry logic and idempotency guarantees.
- **Cross-functional delivery and stakeholder management** - led projects spanning engineering, product, compliance, operations, and finance stakeholders across 120+ Agile sprints; wrote design docs and architecture decisions; mentored 5+ engineers; 300+ peer code reviews; 200+ SQL optimizations (30% performance gain); GitHub Actions CI/CD; incident response with RCA and runbooks.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built income limits import pipeline (1.6M HUD/Tax Credit records); property management and compliance modules (Python, PostgreSQL); AWS S3 cloud migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: AI-Enabled Workflow - Housing Compliance & Renter Eligibility

Agentic Workflow · PropTech · Compliance · Pilot-to-Production · LangGraph

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Claude API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI-enabled workflow automating the renter eligibility investigation process that multifamily operators and compliance teams run manually - the same process TheGuarantors' AI underwriting handles at scale. **Solution architecture:** 8-agent LangGraph orchestration (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) integrating document intelligence, regulatory rules, historical incident data, and live system state into a unified investigation workflow. **Auditability:** every agent decision is traced and citation-backed; HITL guardrails block automated resolution of regulated eligibility paths - LLM reasons, deterministic logic decides regulated outcomes. **Monitoring:** Phase 3 investigation traces false-ineligibility to misconfigured system setting - the multi-source root cause analysis that manual review misses. 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe: AI Monitoring, Auditability & Operational Risk Platform

Monitoring · Auditability · Data Quality · CI Gate · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Claude API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production evaluation and monitoring platform ensuring AI solutions stay reliable, auditable, and compliant in regulated environments. **Quality engine:** 5-metric evaluation (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k variance testing across non-deterministic outputs. **Auditability:** structured tracing of every AI decision with citation chains and confidence scores - the audit trail regulated environments require. **Operational risk controls:** CI-integrated regression gate blocks quality drift before deployment; multi-provider routing (Claude, OpenAI, Ollama) with cost and latency tradeoffs. Open-source, enterprise-deployable. Zero regressions across 1,240+ test cases.

AI on FHIR: AI Solution Deployment - Regulated Healthcare Data

Solutions Architecture · HIPAA · Vendor Integration · Audit Trail · AWS

Python · Flask · React.js · TypeScript · Docker · AWS · BERT · spaCy · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI solution designed and deployed for a regulated healthcare environment - the same architecture pattern required for AI-enabled underwriting and claims workflows. LLM-powered tool invocation translates natural-language queries into structured API calls against FHIR healthcare data; HIPAA-compliant RBAC with authentication and full data isolation; complete audit trail enabling every AI decision to be reviewed; vendor API integration with third-party health data systems; ~60% reduction in manual retrieval time. Demonstrates secure AI solution design, regulatory compliance, and production deployment in a highly regulated domain.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017